

Qualification Pack



Water Technology

QP Code: SGJ/Q4009

Version: 1.0

NSQF Level: 4.5

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Qualification Pack

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Qualification Pack

SGJ/Q4009: Water Technology

Brief Job Description

The individual on the qualification would be specialized to observe, identify, report, maintain, repair equipment and control processes at the water treatment plant and across the networks. The individual would work within sewer networks, for wastewater and sludge treatment in local or industrial wastewater treatment plants. He/She would carry out their work independently based on technical documents, regulations, and legal requirements. They acquire information, plan, and coordinate their work.

Personal Attributes

This job requires the individual to concentrate on the job at hand and complete it without any accidents so diligence and hardworking are desired attributes for individuals performing this role. He/She must also demonstrate strong work ethics, an ability to communicate courteously with co-workers and must be good with following instructions of the supervisor. He/She also has to monitor the activities of Wastewater Treatment Plant and troubleshoot accordingly, so he should be a quick decision maker and good in analysis

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SGJ/N4060: Introduce water treatment fundamentals](#)
2. [SGJ/N4061: Identify components of wastewater treatment plant](#)
3. [SGJ/N4062: Assess water quality](#)
4. [SGJ/N4063: Operate and maintain water treatment plants](#)
5. [SGJ/N4064: Identify and use tools and tackles used in water treatment facility](#)
6. [SGJ/N4065: Explain environment protection regulations and sustainable practices in wastewater treatment](#)
7. [SGJ/N4053: Perform health and safety measures in water treatment plant](#)
8. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Green Jobs
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Sub-Sector	Water Management
Occupation	Technician
Country	India
NSQF Level	4.5
Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/ 3112.9900
Minimum Educational Qualification & Experience	Completed 1st year of UG (UG Certificate) with NA of experience OR 12th Class with 1 Year of experience concerned sector like water management OR 10th grade pass (equivalent) with 1 Year of experience concerned sector like water management OR Certificate-NSQF (4) with 1-2 Years of experience concerned sector like water management
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	16 Years
Last Reviewed On	NA
Next Review Date	30/06/2026
NSQC Approval Date	31/01/2024
Version	1.0
Reference code on NQR	QG-4.5-ES-01816-2024-V1-SCGJ
NQR Version	1.0

Remarks:

Total 510 Hours: (Theory with employability skills: 245 hours+Practical:205 hours+60 hours of On-the-Job Training (OJT))

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SGJ/N4060: Introduce water treatment fundamentals

Description

This unit is about introduction to water treatment fundamentals

Scope

The scope covers the following :

- Basics of water treatment processes
- Sources of Waste water

Elements and Performance Criteria

Basics of water treatment processes

To be competent, the user/individual on the job must be able to:

- PC1.** explain Overview of the water cycle and sources of wastewater and depict water cycle and sources of wastewater through pictures or videos
- PC2.** discuss basics of Water Chemistry
- PC3.** explain Objectives of water treatment
- PC4.** explain parameters for water quality assessment
- PC5.** discuss chemical properties of water
- PC6.** discuss demineralization flow diagram, cation and anion exchangers mixed bed, regeneration of cation and anion exchangers and degasser and illustrate them through pictures or videos
- PC7.** explain basic principles of electricity and illustrate with electrical control of machines and actuators
- PC8.** discuss roles and responsibilities of a water treatment plant technician and prepare a list of roles and responsibilities
- PC9.** discuss functions and objectives of a water treatment plant and showcase the stages of treatment along with flow diagram, schematic and layout of wastewater treatment plant
- PC10.** discuss importance of wastewater treatment for environmental and public health protection
- PC11.** give an overview of global water challenges
- PC12.** discuss historical perspective and evolution of water treatment

Sources of Waste water

To be competent, the user/individual on the job must be able to:

- PC13.** provide an overview of domestic wastewater generation
- PC14.** discuss industrial wastewater generation and its diversity
- PC15.** discuss types of industries producing wastewater (chemical, food processing, textile, etc.) and document the types and volumes of wastewater generated from these locations
- PC16.** discuss characteristics of industrial wastewater: pollutants, toxic substances, pH variations, and temperature
- PC17.** discuss industrial processes contributing to wastewater generation and show how residential wastewater is collected, treated, and disposed of in on-site septic systems

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- PC18.** discuss agricultural wastewater sources: irrigation, livestock farming, and crop processing
- PC19.** organize field trips to different sources of wastewater such as residential areas, industrial sites, agricultural fields, and commercial establishments
- PC20.** construct small models representing residential areas with different wastewater treatment systems (septic tanks, centralized sewer systems)

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** principles and applications of safe working in general and for water and wastewater treatment and operation in the networks and in solid waste management
- KU2.** environmental and safety principles and their application to good housekeeping in the work environment
- KU3.** principles and methods for work organization, control, and management
- KU4.** principles of team working and their applications
- KU5.** the technical language associated with the occupation and the industry

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read English and/or vernacular language
- GS2.** read and understand manuals, health and safety instructions, memos, other company documents
- GS3.** read color codes, as per standard electrical, mechanical and civil nomenclature
- GS4.** express statements or information clearly so that others can hear and understand
- GS5.** participate in and understand the main points of simple discussions
- GS6.** respond appropriately to any queries
- GS7.** communicate with supervisor
- GS8.** follow organization rule-based decision-making process
- GS9.** take decision with systematic course of actions and/or response

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Basics of water treatment processes</i>	19	13	-	-
PC1. explain Overview of the water cycle and sources of wastewater and depict water cycle and sources of wastewater through pictures or videos	1	2	-	-
PC2. discuss basics of Water Chemistry	2	-	-	-
PC3. explain Objectives of water treatment	2	-	-	-
PC4. explain parameters for water quality assessment	2	-	-	-
PC5. discuss chemical properties of water	1	-	-	-
PC6. discuss demineralization flow diagram, cation and anion exchangers mixed bed, regeneration of cation and anion exchangers and degasser and illustrate them through pictures or videos	1	3	-	-
PC7. explain basic principles of electricity and illustrate with electrical control of machines and actuators	1	2	-	-
PC8. discuss roles and responsibilities of a water treatment plant technician and prepare a list of roles and responsibilities	1	3	-	-
PC9. discuss functions and objectives of a water treatment plant and showcase the stages of treatment along with flow diagram, schematic and layout of wastewater treatment plant	2	3	-	-
PC10. discuss importance of wastewater treatment for environmental and public health protection	2	-	-	-
PC11. give an overview of global water challenges	2	-	-	-
PC12. discuss historical perspective and evolution of water treatment	2	-	-	-
<i>Sources of Waste water</i>	9	9	-	-
PC13. provide an overview of domestic wastewater generation	1	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. discuss industrial wastewater generation and its diversity	2	-	-	-
PC15. discuss types of industries producing wastewater (chemical, food processing, textile, etc.) and document the types and volumes of wastewater generated from these locations	1	2	-	-
PC16. discuss characteristics of industrial wastewater: pollutants, toxic substances, pH variations, and temperature	2	-	-	-
PC17. discuss industrial processes contributing to wastewater generation and show how residential wastewater is collected, treated, and disposed of in on-site septic systems	1	3	-	-
PC18. discuss agricultural wastewater sources: irrigation, livestock farming, and crop processing	2	-	-	-
PC19. organize field trips to different sources of wastewater such as residential areas, industrial sites, agricultural fields, and commercial establishments	-	2	-	-
PC20. construct small models representing residential areas with different wastewater treatment systems (septic tanks, centralized sewer systems)	-	2	-	-
NOS Total	28	22	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4060
NOS Name	Introduce water treatment fundamentals
Sector	Green Jobs
Sub-Sector	Water Management
Occupation	Technician
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2026
NSQC Clearance Date	31/01/2024

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SGJ/N4061: Identify components of wastewater treatment plant

Description

This unit is about Wastewater treatment plant and its components

Scope

The scope covers the following :

- Introduce Wastewater Treatment Plant Components

Elements and Performance Criteria

Introduce Wastewater Treatment Plant Components

To be competent, the user/individual on the job must be able to:

- PC1.** discuss water cycle and its components and create scaled-down small models or diagrams representing various treatment plant components (e.g., primary clarifiers, aeration tanks, filtration systems)
- PC2.** discuss groundwater and surface water interactions
- PC3.** discuss and show materials required as metals, composites, plastics, etc. required for wastewater treatment
- PC4.** discuss the basics of mechanical engineering (mechanics, sealing methods, gear technology, etc.)
- PC5.** explain the primary treatment components and demonstrate the operation of primary and secondary clarifiers
- PC6.** discuss types of filtration equipment and their application
- PC7.** explain the Biological Treatment Systems and depict through pictures or Videos
- PC8.** explain Tertiary Treatment Components
- PC9.** discuss chemical addition processes like coagulation and disinfection
- PC10.** discuss the Control and Automation Systems required in wastewater treatment
- PC11.** set up a model demonstrating the aeration process
- PC12.** show working of various Control and Automation Systems required in waste water treatment

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** principles and applications of safe working in general and for water and wastewater treatment and operation in the networks and in solid waste management
- KU2.** the purposes, uses, care, calibration and maintenance of all equipment and materials, together with their safety implications
- KU3.** environmental and safety principles and their application to good housekeeping in the work environment
- KU4.** principles and methods for work organization, control, and management

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- KU5.** principles of team working and their applications
- KU6.** the parameters within which activities need to be scheduled
- KU7.** the technical language associated with the occupation and the industry
- KU8.** the basics of connection technology
- KU9.** the logical sequence of network flow and purification steps

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and understand manuals, health and safety instructions, memos, other company documents
- GS2.** read color codes, as per standard electrical, mechanical and civil nomenclature
- GS3.** participate in and understand the main points of simple discussions
- GS4.** respond appropriately to any queries
- GS5.** follow organization rule-based decision-making process
- GS6.** communicate with supervisor
- GS7.** plan and organize of work to meet deadlines
- GS8.** apply domain knowledge, observations and data to select course of action to perform tasks related to wastewater treatment
- GS9.** approach relevant authority when required

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduce Wastewater Treatment Plant Components</i>	23	27	-	-
PC1. discuss water cycle and its components and create scaled-down small models or diagrams representing various treatment plant components (e.g., primary clarifiers, aeration tanks, filtration systems)	2	4	-	-
PC2. discuss groundwater and surface water interactions	2	-	-	-
PC3. discuss and show materials required as metals, composites, plastics, etc. required for wastewater treatment	2	5	-	-
PC4. discuss the basics of mechanical engineering (mechanics, sealing methods, gear technology, etc.)	2	-	-	-
PC5. explain the primary treatment components and demonstrate the operation of primary and secondary clarifiers	2	3	-	-
PC6. discuss types of filtration equipment and their application	2	-	-	-
PC7. explain the Biological Treatment Systems and depict through pictures or Videos	2	5	-	-
PC8. explain Tertiary Treatment Components	3	-	-	-
PC9. discuss chemical addition processes like coagulation and disinfection	3	-	-	-
PC10. discuss the Control and Automation Systems required in wastewater treatment	3	-	-	-
PC11. set up a model demonstrating the aeration process	-	5	-	-
PC12. show working of various Control and Automation Systems required in waste water treatment	-	5	-	-
NOS Total	23	27	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4061
NOS Name	Identify components of wastewater treatment plant
Sector	Green Jobs
Sub-Sector	Water Management
Occupation	Technician
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2026
NSQC Clearance Date	31/01/2024

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SGJ/N4062: Assess water quality

Description

This unit is about Water Quality Assessment

Scope

The scope covers the following :

- Sampling techniques and testing methods for water treatment

Elements and Performance Criteria

Sampling techniques and testing Methods for water treatment

To be competent, the user/individual on the job must be able to:

- PC1.** explain grab sampling and demonstrate techniques using sample bottles or containers
- PC2.** discuss principles and procedures of grab sampling and show how to mix and create a composite sample for analysis
- PC3.** discuss types of grab samples
- PC4.** discuss Parameters for assessing water quality
- PC5.** discuss drinking water treatment methods like filtration, disinfection and purification
- PC6.** discuss water quality standards and Regulations
- PC7.** discuss the process of membrane preparation
- PC8.** explain sample container selection and preparation and Set up small automated sampling equipment (if available) and showcase its operation.
- PC9.** discuss the importance of proper labeling and documentation
- PC10.** explain physical testing methods
- PC11.** discuss methods and instruments for physical parameter analysis and demonstrate different methods for sample preservation, such as adding preservatives or controlling temperature, to maintain sample integrity during transportation and storage
- PC12.** explain laboratory techniques and equipment for chemical parameter analysis
- PC13.** discuss Advanced Water Treatment Technologies like Membrane filtration (Reverse Osmosis, Ultrafiltration), Advanced oxidation processes, Desalination technologies
- PC14.** discuss microbiological testing methods: membrane filtration, multiple tube fermentation, and other techniques
- PC15.** discuss metals and toxic substances testing
- PC16.** explain variations in sampling techniques at different stages of treatment
- PC17.** perform tests for physical parameters like temperature, turbidity, and color using appropriate meters or instruments
- PC18.** conduct chemical tests for parameters like pH, dissolved oxygen, and chemical oxygen demand (COD)

Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** the purposes, uses, care, calibration and maintenance of all equipment and materials, together with their safety implications
- KU2.** principles and methods for work organization, control, and management
- KU3.** the technical language associated with the occupation and the industry
- KU4.** the standards required for routine and exception reporting in oral, written, and electronic form (e.g., values, figures, units, minimal information, recommendations)
- KU5.** the purposes and techniques for generating, maintaining, and presenting records
- KU6.** the logical sequence of network flow and purification steps
- KU7.** the basic calculations required within water and wastewater network and treatment processes

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read English and/or vernacular language
- GS2.** read and understand manuals, health and safety instructions, memos, other company documents
- GS3.** read from different sources- books screens in machines and signage
- GS4.** read color codes, as per standard electrical, mechanical and civil nomenclature
- GS5.** express statements or information clearly so that others can hear and understand
- GS6.** participate in and understand the main points of simple discussions
- GS7.** follow organization rule-based decision-making process
- GS8.** plan and organize of work to meet deadlines
- GS9.** work constructively and collaboratively with others
- GS10.** choose best methods to complete assigned tasks
- GS11.** apply domain knowledge, observations and data to select course of action to perform tasks related to Wastewater Treatment
- GS12.** ask questions for better understanding

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Sampling techniques and testing Methods for water treatment</i>	25	25	-	-
PC1. explain grab sampling and demonstrate techniques using sample bottles or containers	1	2	-	-
PC2. discuss principles and procedures of grab sampling and show how to mix and create a composite sample for analysis	1	2	-	-
PC3. discuss types of grab samples	2	-	-	-
PC4. discuss Parameters for assessing water quality	2	-	-	-
PC5. discuss drinking water treatment methods like filtration, disinfection and purification	2	-	-	-
PC6. discuss water quality standards and Regulations	2	-	-	-
PC7. discuss the process of membrane preparation	2	-	-	-
PC8. explain sample container selection and preparation and Set up small automated sampling equipment (if available) and showcase its operation.	1	4	-	-
PC9. discuss the importance of proper labeling and documentation	1	-	-	-
PC10. explain physical testing methods	1	-	-	-
PC11. discuss methods and instruments for physical parameter analysis and demonstrate different methods for sample preservation, such as adding preservatives or controlling temperature, to maintain sample integrity during transportation and storage	2	4	-	-
PC12. explain laboratory techniques and equipment for chemical parameter analysis	2	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. discuss Advanced Water Treatment Technologies like Membrane filtration (Reverse Osmosis, Ultrafiltration), Advanced oxidation processes, Desalination technologies	2	-	-	-
PC14. discuss microbiological testing methods: membrane filtration, multiple tube fermentation, and other techniques	2	-	-	-
PC15. discuss metals and toxic substances testing	2	-	-	-
PC16. explain variations in sampling techniques at different stages of treatment	-	4	-	-
PC17. perform tests for physical parameters like temperature, turbidity, and color using appropriate meters or instruments	-	4	-	-
PC18. conduct chemical tests for parameters like pH, dissolved oxygen, and chemical oxygen demand (COD)	-	5	-	-
NOS Total	25	25	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4062
NOS Name	Assess water quality
Sector	Green Jobs
Sub-Sector	Water Management
Occupation	Technician
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2026
NSQC Clearance Date	31/01/2024

Qualification Pack

SGJ/N4063: Operate and maintain water treatment plants

Description

This unit is about operations and maintenance of water Treatment Plant

Scope

The scope covers the following :

- Operation and Maintenance of water Treatment Plant

Elements and Performance Criteria

Operation and Maintenance of water Treatment Plant

To be competent, the user/individual on the job must be able to:

- PC1.** explain protocols for starting up and shutting down treatment processes
- PC2.** discuss water supply systems and networks
- PC3.** discuss overview of process control systems and their importance
- PC4.** discuss about the monitoring of parameters for different treatment stages (pH, flow rates, etc.)
- PC5.** discuss and monitor operating conditions, record instruments and gauges reading/operation data in the daily log sheets and maintain records
- PC6.** explain daily, weekly and monthly checks list for equipment and process performance
- PC7.** discuss O&M and troubleshooting of pumps & valves and conduct demonstrations to operate various equipment used in wastewater treatment plants, such as pumps, blowers, mixers, and valves
- PC8.** discuss inspection and maintenance of conveyance systems (pipelines, channels) and show cleaning, lubrication and inspection procedures
- PC9.** discuss the basic principles of sensor technology
- PC10.** discuss basic calculations required within water and wastewater treatment network
- PC11.** discuss the basic principles and functionality of closed loop technology
- PC12.** discuss the basic principles of actuators
- PC13.** discuss the water loss, leakage and its potential causes with solutions for prevention
- PC14.** explain preventive and corrective maintenance practices for filters, tanks, gauges, pump, and valves, etc. in waste water treatment in time bound manner
- PC15.** discuss recording and documentation of operational data
- PC16.** demonstrate circuit, P&I-diagrams, operating manuals and instruction manuals
- PC17.** operate the ion exchangers viz. anion exchanger, cation exchanger and mixed bed exchanger
- PC18.** showcase the operation of pumps used in wastewater treatment
- PC19.** prepare standard procedures for startup, shutdown of waste water treatment plant
- PC20.** demonstrate maintenance tasks of treatment components like clarifiers, aeration tanks, filters, and disinfection systems

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** principles and applications of safe working in general and for water and wastewater treatment and operation in the networks and in solid waste management
- KU2.** the purposes, uses, care, calibration and maintenance of all equipment and materials, together with their safety implications
- KU3.** environmental and safety principles and their application to good housekeeping in the work environment
- KU4.** principles and methods for work organization, control, and management
- KU5.** principles of team working and their applications
- KU6.** the technical language associated with the occupation and the industry
- KU7.** the purposes and techniques for generating, maintaining, and presenting records
- KU8.** techniques and options for making mechanical repairs
- KU9.** sustainable practices which support human, ecological, economic health and vitality
- KU10.** the basics and principles of sample pre-treatment, storage, sample preserving and sample collection

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read English and/or vernacular language
- GS2.** read and understand manuals, health and safety instructions, memos, other company documents
- GS3.** read from different sources- books screens in machines and signage
- GS4.** read color codes, as per standard electrical, mechanical and civil nomenclature
- GS5.** express statements or information clearly so that others can hear and understand
- GS6.** follow organization rule-based decision-making process
- GS7.** take decision with systematic course of actions and/or response
- GS8.** evaluate information obtained from customers, supervisor and co-workers to perform day to day activities
- GS9.** apply domain knowledge, observations and data to select course of action to perform tasks related to Wastewater Treatment
- GS10.** ask questions for better understanding

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Operation and Maintenance of water Treatment Plant</i>	21	29	-	-
PC1. explain protocols for starting up and shutting down treatment processes	2	-	-	-
PC2. discuss water supply systems and networks	2	-	-	-
PC3. discuss overview of process control systems and their importance	2	-	-	-
PC4. discuss about the monitoring of parameters for different treatment stages (pH, flow rates, etc.)	2	-	-	-
PC5. discuss and monitor operating conditions, record instruments and gauges reading/operation data in the daily log sheets and maintain records	2	-	-	-
PC6. explain daily, weekly and monthly checks list for equipment and process performance	2	-	-	-
PC7. discuss O&M and troubleshooting of pumps & valves and conduct demonstrations to operate various equipment used in wastewater treatment plants, such as pumps, blowers, mixers, and valves	1	4	-	-
PC8. discuss inspection and maintenance of conveyance systems (pipelines, channels) and show cleaning, lubrication and inspection procedures	1	4	-	-
PC9. discuss the basic principles of sensor technology	1	-	-	-
PC10. discuss basic calculations required within water and wastewater treatment network	1	-	-	-
PC11. discuss the basic principles and functionality of closed loop technology	1	-	-	-
PC12. discuss the basic principles of actuators	1	-	-	-
PC13. discuss the water loss, leakage and its potential causes with solutions for prevention	1	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. explain preventive and corrective maintenance practices for filters, tanks, gauges, pump, and valves, etc. in waste water treatment in time bound manner	1	4	-	-
PC15. discuss recording and documentation of operational data	1	-	-	-
PC16. demonstrate circuit, P&I-diagrams, operating manuals and instruction manuals	-	4	-	-
PC17. operate the ion exchangers viz. anion exchanger, cation exchanger and mixed bed exchanger	-	4	-	-
PC18. showcase the operation of pumps used in wastewater treatment	-	4	-	-
PC19. prepare standard procedures for startup, shutdown of waste water treatment plant	-	3	-	-
PC20. demonstrate maintenance tasks of treatment components like clarifiers, aeration tanks, filters, and disinfection systems	-	2	-	-
NOS Total	21	29	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4063
NOS Name	Operate and maintain water treatment plants
Sector	Green Jobs
Sub-Sector	Water Management
Occupation	Technician
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2026
NSQC Clearance Date	31/01/2024

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SGJ/N4064: Identify and use tools and tackles used in water treatment facility

Description

This unit is about tools and tackles used in water treatment facility

Scope

The scope covers the following :

- Tools and tackles used in water treatment facility

Elements and Performance Criteria

Tools and tackles used in water treatment facility

To be competent, the user/individual on the job must be able to:

- PC1.** discuss basic hand tools used in wastewater treatment and demonstrate the use of essential hand tools such as wrenches, screwdrivers, pliers, hammers, and saws
- PC2.** discuss about the measuring and testing instruments used in wastewater treatment
- PC3.** discuss about power tools such as drills, grinders and sanders and machinery used in wastewater treatment and showcase their usages
- PC4.** discuss and demonstrate pipe cutting and welding tools and equipment used in wastewater treatment
- PC5.** discuss about cleaning and maintenance tools used in wastewater treatment and demonstrate proper lubrication techniques using grease guns, oil dispensers, and lubrication systems
- PC6.** discuss about lubrication tools and equipment used in wastewater treatment
- PC7.** discuss about tool maintenance and storage practices
- PC8.** discuss checking and detection of damaged tools
- PC9.** explain the importance of tool maintenance and regular inspections
- PC10.** demonstrate the calibration and use of measuring instruments like pressure gauges, flow meters, pH meters, and conductivity meters

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** principles and applications of safe working in general and for water and wastewater treatment and operation in the networks and in solid waste management
- KU2.** environmental and safety principles and their application to good housekeeping in the work environment
- KU3.** principles and methods for work organization, control, and management
- KU4.** principles of team working and their applications
- KU5.** the range and purposes of documentation in both paper and electronic form

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- KU6.** the standards required for routine and exception reporting in oral, written, and electronic form (e.g., values, figures, units, minimal information, recommendations)
- KU7.** the dangers/hazards of electrical systems
- KU8.** the basics of electrical control of machines and actuators
- KU9.** techniques and options for making mechanical repairs
- KU10.** the basics of mechanical engineering (mechanics, sealing methods, gear technology, etc.)
- KU11.** the proper use of each specific glassware, analytical equipment or instrument
- KU12.** the meaning of relevant danger and safety symbols/signage

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read English and/or vernacular language
- GS2.** read and understand manuals, health and safety instructions, memos, other company documents
- GS3.** read color codes, as per standard electrical, mechanical and civil nomenclature
- GS4.** express statements or information clearly so that others can hear and understand
- GS5.** participate in and understand the main points of simple discussions
- GS6.** take decision with systematic course of actions and/or response
- GS7.** apply domain knowledge, observations and data to select course of action to perform tasks related to wastewater treatment
- GS8.** evaluate information obtained from customers, supervisor and co-workers to perform day to day activities
- GS9.** plan and organize of work to meet deadlines
- GS10.** ask questions for better understanding

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Tools and tackles used in water treatment facility</i>	25	25	-	-
PC1. discuss basic hand tools used in wastewater treatment and demonstrate the use of essential hand tools such as wrenches, screwdrivers, pliers, hammers, and saws	2	5	-	-
PC2. discuss about the measuring and testing instruments used in wastewater treatment	4	-	-	-
PC3. discuss about power tools such as drills, grinders and sanders and machinery used in wastewater treatment and showcase their usages	2	5	-	-
PC4. discuss and demonstrate pipe cutting and welding tools and equipment used in wastewater treatment	2	5	-	-
PC5. discuss about cleaning and maintenance tools used in wastewater treatment and demonstrate proper lubrication techniques using grease guns, oil dispensers, and lubrication systems	2	5	-	-
PC6. discuss about lubrication tools and equipment used in wastewater treatment	3	-	-	-
PC7. discuss about tool maintenance and storage practices	3	-	-	-
PC8. discuss checking and detection of damaged tools	3	-	-	-
PC9. explain the importance of tool maintenance and regular inspections	4	-	-	-
PC10. demonstrate the calibration and use of measuring instruments like pressure gauges, flow meters, pH meters, and conductivity meters	-	5	-	-
NOS Total	25	25	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4064
NOS Name	Identify and use tools and tackles used in water treatment facility
Sector	Green Jobs
Sub-Sector	Water Management
Occupation	Technician
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2026
NSQC Clearance Date	31/01/2024

Qualification Pack

SGJ/N4065: Explain environment protection regulations and sustainable practices in wastewater treatment

Description

This unit is about environment protection regulations and sustainable practices in wastewater treatment

Scope

The scope covers the following :

- Overview of environmental regulations and sustainable practices in wastewater treatment

Elements and Performance Criteria

Overview of environmental regulations and sustainable practices in wastewater treatment

To be competent, the user/individual on the job must be able to:

- PC1.** discuss and show the environmental impact of untreated wastewater through pictures or videos
- PC2.** explain the impact of wastewater discharge on aquatic ecosystems
- PC3.** discuss water conservation strategies and reduce, reuse, and recycle of water resources
- PC4.** explain the methods and tools used in assessing environmental impacts
- PC5.** give an overview of local, national and international regulations and present case studies or real-world examples of past regulatory violations related to wastewater
- PC6.** discuss and showcase list of eco-friendly treatment methods for treating wastewater treatment
- PC7.** discuss rainwater harvesting and storm water management
- PC8.** discuss the concept of zero liquid discharge
- PC9.** discuss and showcase the successful examples of environmentally friendly wastewater treatment projects
- PC10.** discuss challenges in adopting sustainable wastewater treatment practices
- PC11.** explain the importance of environmental protection and sustainability in wastewater treatment

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** principles and applications of safe working in general and for water and wastewater treatment and operation in the networks and in solid waste management
- KU2.** the purposes, uses, care, calibration and maintenance of all equipment and materials, together with their safety implications
- KU3.** environmental and safety principles and their application to good housekeeping in the work environment
- KU4.** principles and methods for work organization, control, and management

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- KU5.** the purposes and techniques for generating, maintaining, and presenting records
- KU6.** sustainable practices which support human, ecological, economic health and vitality
- KU7.** basics principles and practices of hygiene
- KU8.** the hazardous aspects/points for the environment (danger/risk analysis)
- KU9.** basic operation/function of laboratory equipment
- KU10.** analytical techniques for fault finding and solving

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read English and/or vernacular language
- GS2.** read and understand manuals, health and safety instructions, memos, other company documents
- GS3.** express statements or information clearly so that others can hear and understand
- GS4.** read color codes, as per standard electrical, mechanical and civil nomenclature
- GS5.** follow organization rule-based decision-making process
- GS6.** plan and organize of work to meet deadlines
- GS7.** recognize problems and search for solutions
- GS8.** choose best methods to complete assigned tasks
- GS9.** approach relevant authority when required
- GS10.** evaluate information obtained from customers, supervisor and co-workers to perform day to day activities

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Overview of environmental regulations and sustainable practices in wastewater treatment</i>	30	20	-	-
PC1. discuss and show the environmental impact of untreated wastewater through pictures or videos	2	5	-	-
PC2. explain the impact of wastewater discharge on aquatic ecosystems	3	-	-	-
PC3. discuss water conservation strategies and reduce, reuse, and recycle of water resources	3	-	-	-
PC4. explain the methods and tools used in assessing environmental impacts	3	-	-	-
PC5. give an overview of local, national and international regulations and present case studies or real-world examples of past regulatory violations related to wastewater	2	5	-	-
PC6. discuss and showcase list of eco-friendly treatment methods for treating wastewater treatment	2	5	-	-
PC7. discuss rainwater harvesting and storm water management	3	-	-	-
PC8. discuss the concept of zero liquid discharge	4	-	-	-
PC9. discuss and showcase the successful examples of environmentally friendly wastewater treatment projects	2	5	-	-
PC10. discuss challenges in adopting sustainable wastewater treatment practices	3	-	-	-
PC11. explain the importance of environmental protection and sustainability in wastewater treatment	3	-	-	-
NOS Total	30	20	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4065
NOS Name	Explain environment protection regulations and sustainable practices in wastewater treatment
Sector	Green Jobs
Sub-Sector	Water Management
Occupation	Technician
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2026
NSQC Clearance Date	31/01/2024

Qualification Pack

SGJ/N4053: Perform health and safety measures in water treatment plant

Description

This unit is about to perform health and safety measures in water treatment plant

Scope

The scope covers the following :

- Perform health and safety measures in water treatment plant

Elements and Performance Criteria

Perform health and safety measures in water treatment plant

To be competent, the user/individual on the job must be able to:

- PC1.** explain the requirements for safe work area at wastewater treatment plant
- PC2.** explain the importance of occupational health & Safety standards and regulations
- PC3.** discuss importance of different detectors and safety tools used in wastewater treatment plant
- PC4.** review the Material Safety Data Sheet(MSDS) and labels of chemicals contained in cylinders in order to be aware of their hazards and precautionary measures
- PC5.** explain the importance of administering first aid and demonstrate how to administer first aid
- PC6.** identify and use the personal protective equipment used for the specific purpose
- PC7.** identify potential hazards (chemical, biological, physical) in wastewater treatment plants and demonstrate how to follow necessary and adequate safety measures
- PC8.** discuss mock testing of firefighting system and demonstrate the use of fire extinguishers, fire detection and alarm system
- PC9.** explain the meaning of relevant danger and safety symbols/signage
- PC10.** discuss all applicable statutory requirements along with safety regulations in terms of fire protection
- PC11.** discuss basics principles and practices of hygiene
- PC12.** incorporate good housekeeping practices and infection control & prevention guidelines
- PC13.** explain reporting procedures for accidents, potential hazards, near misses, permit to work and show how to comply with all applicable statutory requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** principles and applications of safe working in general and for water and wastewater treatment and operation in the networks and in solid waste management
- KU2.** the purposes, uses, care, calibration and maintenance of all equipment and materials, together with their safety implications

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- KU3.** environmental and safety principles and their application to good housekeeping in the work environment
- KU4.** principles and methods for work organization, control, and management
- KU5.** the standards required for routine and exception reporting in oral, written, and electronic form (e.g., values, figures, units, minimal information, recommendations)
- KU6.** the dangers/hazards of electrical systems
- KU7.** sustainable practices which support human, ecological, economic health and vitality
- KU8.** risk assessment for (biological, chemical, electrical, thermal and mechanical-operations)
- KU9.** health and work-related regulations
- KU10.** health maintaining regulations, personal protection equipment (PPE)

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read English and/or vernacular language
- GS2.** read and understand manuals, health and safety instructions, memos, other company documents
- GS3.** read color codes, as per standard electrical, mechanical and civil nomenclature
- GS4.** express statements or information clearly so that others can hear and understand
- GS5.** respond appropriately to any queries
- GS6.** work constructively and collaboratively with others
- GS7.** recognize problems and search for solutions
- GS8.** evaluate information obtained from customers, supervisor and co-workers to perform day to day activities
- GS9.** ask questions for better understanding

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform health and safety measures in water treatment plant</i>	25	25	-	-
PC1. explain the requirements for safe work area at wastewater treatment plant	3	-	-	-
PC2. explain the importance of occupational health & Safety standards and regulations	2	-	-	-
PC3. discuss importance of different detectors and safety tools used in wastewater treatment plant	2	-	-	-
PC4. review the Material Safety Data Sheet(MSDS) and labels of chemicals contained in cylinders in order to be aware of their hazards and precautionary measures	2	-	-	-
PC5. explain the importance of administering first aid and demonstrate how to administer first aid	2	5	-	-
PC6. identify and use the personal protective equipment used for the specific purpose	2	5	-	-
PC7. identify potential hazards (chemical, biological, physical) in wastewater treatment plants and demonstrate how to follow necessary and adequate safety measures	1	5	-	-
PC8. discuss mock testing of firefighting system and demonstrate the use of fire extinguishers, fire detection and alarm system	1	4	-	-
PC9. explain the meaning of relevant danger and safety symbols/signage	2	-	-	-
PC10. discuss all applicable statutory requirements along with safety regulations in terms of fire protection	2	-	-	-
PC11. discuss basics principles and practices of hygiene	2	-	-	-
PC12. incorporate good housekeeping practices and infection control & prevention guidelines	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. explain reporting procedures for accidents, potential hazards, near misses, permit to work and show how to comply with all applicable statutory requirements	2	3	-	-
NOS Total	25	25	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4053
NOS Name	Perform health and safety measures in water treatment plant
Sector	Green Jobs
Sub-Sector	Water Management
Occupation	Technician
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2026
NSQC Clearance Date	31/01/2024

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

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Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	22/01/2026
Next Review Date	22/01/2029
NSQC Clearance Date	22/01/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Assessment is governed by the WorldSkills Assessment Strategy. The Strategy establishes the principles and techniques to which WorldSkills assessment and marking must conform. Expert assessment practice lies at the heart of the WorldSkills Competition. For this reason, it is the subject of continuing professional development and scrutiny. The growth of expertise in assessment will inform the future use and direction of the main assessment instruments used by the WorldSkills Competition: the Marking Scheme, Test Project, and Competition Information System (CIS).

Assessment at the WorldSkills Competition falls into two broad types: Measurement and Judgement. For both types of assessment, the use of explicit benchmarks against which to assess each Aspect is essential to guarantee quality.

The Marking Scheme must follow the weightings within the Standards. The Test Project is the assessment vehicle for the skill competition, and therefore also follows the Standards. The CIS enables the timely and accurate recording of marks; its capacity for scrutiny, support, and feedback is continuously expanding.

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The Marking Scheme, in outline, will lead the process of Test Project design. After this, the Marking Scheme and Test Project will be designed, developed, and verified through an iterative process, to ensure that both together optimize their relationship with the Standards and the Assessment Strategy. They will be agreed by the Experts and submitted to WSI for approval together, to demonstrate their quality and conformity with the Standards. Prior to submission for approval to WSI, the Marking Scheme and Test Project will liaise with the WSI Skill Advisors for quality assurance and to benefit from the capabilities of the CIS.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N4060.Introduce water treatment fundamentals	28	22	-	-	50	12
SGJ/N4061.Identify components of wastewater treatment plant	23	27	-	-	50	13
SGJ/N4062.Assess water quality	25	25	-	-	50	12
SGJ/N4063.Operate and maintain water treatment plants	21	29	-	-	50	13
SGJ/N4064.Identify and use tools and tackles used in water treatment facility	25	25	-	-	50	12
SGJ/N4065.Explain environment protection regulations and sustainable practices in wastewater treatment	30	20	-	-	50	13
SGJ/N4053.Perform health and safety measures in water treatment plant	25	25	-	-	50	12

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	13
Total	197	203	-	-	400	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.